NAÏVE BAYES

Naïve Bayes Analysis

- Naive-Bayes (NB) technique is a supervised learning technique that uses probability-theory-based analysis.
- It is a machine-learning technique that computes the probabilities of an instance belonging to each one of many target classes, given the prior probabilities of classification using individual factors.
- Naïve Bayes technique is used often in classifying text documents into one of multiple predefined categories.

- NB algorithm is easy to understand and works fast.
- It also performs well in multiclass prediction, such as when the target class has multiple options beyond binary yes/no classification.
- NB can perform well even in case of categorical input variables compared to numerical variable(s)

- In the abstract, Naive-Bayes is a conditional probability model for classification purposes.
- The goal is to find a way to predict the class variable (Y) using a vector of independent variables i.e., finding the function f: X $\rightarrow$ Y.
- In probability terms, the goal is to find P(Y|X), i.e., the probability of Y belonging to a certain class X.
- Y is generally assumed to be a categorical variable with two or more discrete values.

- Given an instance to be classified, represented by a vector $x = (x_1, x_2 \ldots\ldots x_n)$ representing 'n' features (independent variables), the Naive-Bayes model assigns, to an instance, probabilities of belonging to any of the K classes.
- The class K with the highest posterior probability is the label assigned to the instance.

- The posterior probability (of belonging to a Class K) is calculated as a function of prior probabilities and current likelihood value, as shown in the equation,

$$\boldsymbol{p(c_k|x) = \frac{p(c_k) * p(x|c_k)}{p(x)}}$$

- $p(c_k|x)$ is the posterior probability of class K, given predictor X.
- $p(c_k)$ is the prior probability of class K.
- $p(x)$ is the prior probability of predictor.
- $p(x|c_k)$ is the current likelihood of predictor given class.

Naïve Bayes Model – Simple Classification Example

- Suppose a **XYZ** company needs to predict the service required by the incoming customer.
- If there are only two services offered — R and M.
- Then the value to be predicted is whether the next customer will be for R or M.
- The number of classes k are 2 that is, k=2.

- The first step is to compute prior probability.
- Suppose the data gathered for the last one year showed that during that period there were 2500 customers for R and 1500 customers for M.
- Thus, prior probability for the next customer to be for R is 2500/4000 or 5/8.
- Prior probability for the next customer to be for M is 1500/4000 or 3/8.
- Based on this information alone, the next customer would likely be for R.

- Another way to predict the service requirement by the next customer is to look at the most recent data.
- One can look at the last few (choose a number) customers, to predict the next customer.
- Suppose the last five customers were for the services ... R, M, R, M, M order.
- Thus, the data shows the recent probability of R is 2/5 and that of M is 3/5.
- Based on just this information, the next customer will likely to be for M.

- Thomas Bayes suggested that the prior probability should be informed by the more recent data.
- Naive-Bayes posterior probability for a class is computed by multiplying the prior probability and the recent probability.
- NB posterior probability P(R) is (5/8 x 2/5) = 10/40.
- Similarly, the NB probability P(M) is (3/8 x 3/5) = 9/40.
- Since P(R) is greater than P(M), it follows that there is a greater probability of the next customer to be for R.
- Thus the expected class label assigned to the next customer would be R.

- Suppose, however the next customer coming in was for M service.
- The last five customer sequence now becomes M, R, M, M, M.
- Thus, the recent data shows the probability for R to be 1/5 and that of M to be 4/5.
- Now the NB probability for R is (5/8 x 1/5) = 5/40.
- Similarly, the NB probability for M is (3/8 x 4/5) = 12/40.
- Since P(M) is greater than P(R), it follows that there is a greater probability of the next customer to be for M.
- Thus the expected class label assigned to the next customer is M.

Naïve Bayes Model – Advantages and Disadvantages

- The NB logic is simple and so is the NB posterior probability computation.
- Conditional probabilities can be computed for discrete data and for probabilistic distributions.
- Naive-Bayes assumes that all the features are independent for most instances that work fine. However, it can be a limitation. If there are no joint occurrences at all of a class label with a certain attribute, then the frequency-based conditional probability will be zero.
- A limitation of NB is that the posterior probability computations are good for comparison and classification of the instances. However, the probability values themselves are not good estimates of the event happening.